

Module 4: Cells - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. Which statement is part of cell theory?

- A. Cells arise only from nonliving matter.
- B. All organisms are made of cells, and cells come from existing cells.
- C. Cells are larger than the organisms they build.
- D. Only animals are made of cells.
- Answer: B
- Why: Cell theory states that organisms are made of cells and that cells arise from pre-existing cells.

2. What is the clearest structural difference between a prokaryotic and a eukaryotic cell?

- A. Only prokaryotes contain DNA.
- B. Only eukaryotes are alive.
- C. Prokaryotes are always larger.
- D. Eukaryotes have a membrane-bound nucleus; prokaryotes do not.
- Answer: D
- Why: The defining feature is the membrane-bound nucleus (and other organelles) found in eukaryotes but not prokaryotes.

3. A cell is packed with mitochondria. This most likely means the cell:

- A. has a high demand for usable energy
- B. does not use any energy
- C. performs only photosynthesis
- D. contains no DNA
- Answer: A
- Why: Mitochondria harvest usable energy, so energy-hungry cells such as muscle cells contain many of them.

4. Which structure would you find in a plant cell but not an animal cell?

- A. Nucleus
- B. Mitochondrion
- C. Chloroplast
- D. Cell membrane

◦ Answer: C

◦ Why: Chloroplasts carry out photosynthesis and are found in plants and algae, not in animal cells.